FAULHABER

Encoders

magnetic Encoder, digital outputs,
3 channels, 16 - 4096 lines per revolution



Series IEF3-4096

	IEF3	-16	-32	-64	-128	-256	-512	-1024	-2048	-4096	
Lines per revolution	*N*	16	32	64	128	256	512	1 024	2 048	4 096	
Frequency range, up to[1]	*f*	5	10	20	40	80	160	320	640	875	kHz
Signal output, square wave		2+1 Index									Channels
Supply voltage	U_{DD}	4,5 ... 5,5									V
Current consumption, typical[2]	I_{DD}	typ. 25, max. 40									mA
Output current, max.[3]	I_{OUT}	2,5									mA
Index Pulse width[4]	P_0	90 ± 45						90 ± 65	90 ± 75		°e
Phase shift, channel A to B	Φ	90 ± 45						90 ± 65	90 ± 75		°e
Signal rise/fall time, max. (C_{LOAD} = 50 pF)	*tr*/*tf*	0,05 / 0,05									µs
Inertia of sensor magnet	*J*	1,57									gcm²
Operating temperature range		-40 ... +100									°C
Accuracy, typ.		0,5									°m
Repeatability, typ.		0,08									°m
Hysteresis		0,02									°m
Edge spacing, min.		225									ns
MTTF[5]		6 668 000									h
Mass, typ.		15,4									g

[1] Velocity (min^{-1}) = *f* (Hz) x 60/*N*
[2] U_{DD} = 5 V: with unloaded outputs
[3] U_{DD} = 5 V: low logic level < 0,4 V, high logic level > 4,5 V: CMOS- and TTL compatible
[4] At 5 000 min^{-1}
[5] „Mean time to failure" at maximum operating condition

For combination with Motor

Dimensional drawing A	L1 [mm]
2214 ... BXT H	21,0

Dimensional drawing B	L1 [mm]
3216 ... BXT H	23,0

Dimensional drawing C	L1 [mm]
4221 ... BXT H	28,0

Characteristics

These incremental encoders with 3 output channels, in combination with the FAULHABER Brushless DC-Motors, are used for the indication and control of both shaft velocity and direction of rotation as well as for positioning.

The encoder is integrated in the Brushless DC-Motors BXT H-Series and extends the overall length by only 6,2 mm.

A segmented magnetic disc provides a magnetic field which is detected and further processed by an angle sensor.

At the encoder outputs, two 90° phase-shifted square wave signals are available with up to 4096 impulses and an index impulse per motor revolution.
The encoder is available with different standard resolutions.

The supply voltage for the encoder and the output signals are interfaced through a ribbon cable, optional with connector.

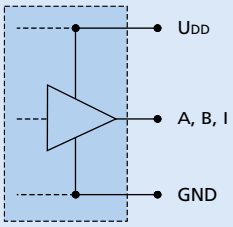
Details for the Brushless DC-Motors and suitable reduction gearheads are on separate catalogue pages.

To view our large range of accessory parts, please refer to the "Accessories" chapter.

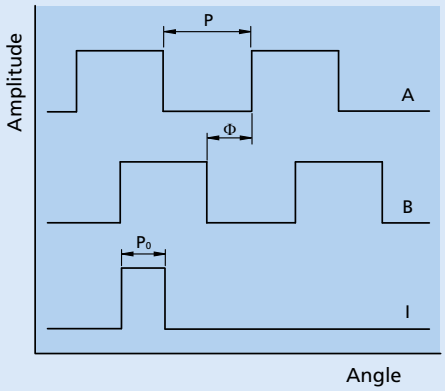
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Circuit diagram / Output signals

Output circuit



Output signals
with clockwise rotation as seen from the shaft end



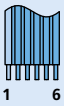
Connector information / Variants

Example product designation: **2214S012BXTH IEF3-4096**

Option	Type	Description
3807	Connector 	for combination with Brushless DC-Flat Motors series BXT H. Connector variants AWG 28 / PVC ribbon cable with connector MOLEX Picoblade 51021-0600, recommended mating connector 53047-0610.
3592	Connector 	for combination with Brushless DC-Flat Motors series BXT H. Connector variants AWG 28 / PVC ribbon cable with connector MOLEX Picoblade 51021-0600, recommended mating connector 53047-0610.  Inclusive motor connector 3830

Connection Encoder

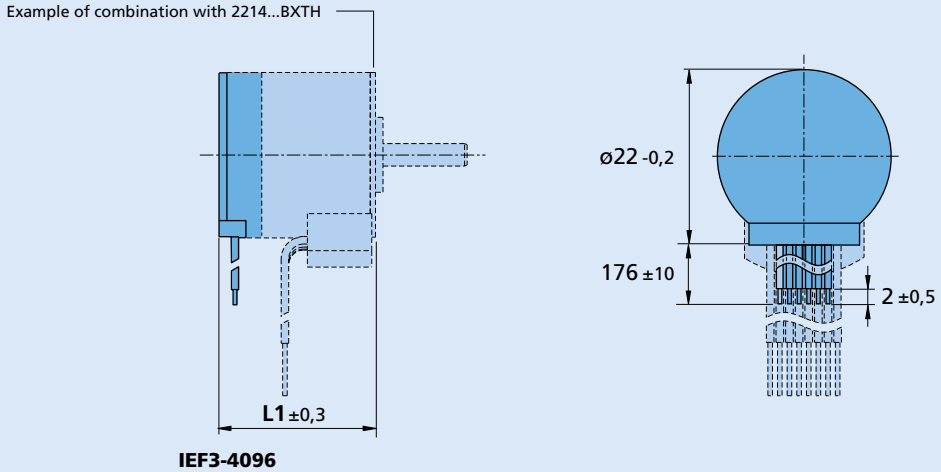
No.	Function
1	N.C.
2	Channel I
3	GND
4	UDD
5	Channel B
6	Channel A



Standard cable
PVC-ribbon cable, 6-AWG 28, 1,27 mm

Caution:
Incorrect lead connection will damage the motor electronics!

Dimensional drawing A

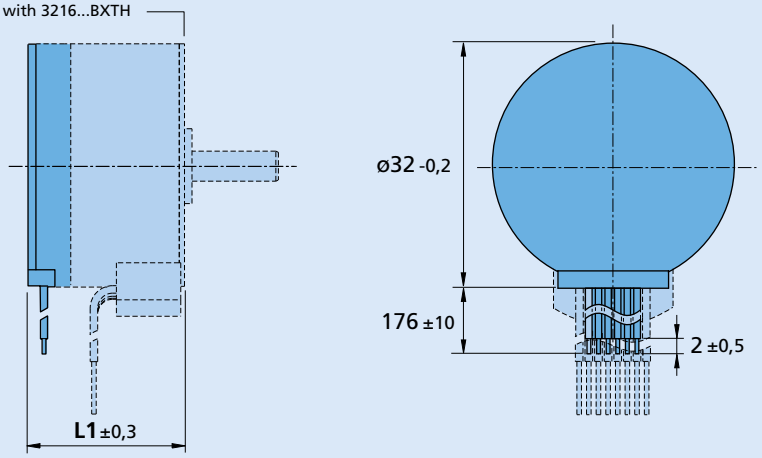


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Dimensional drawing B

Example of combination with 3216...BXTH

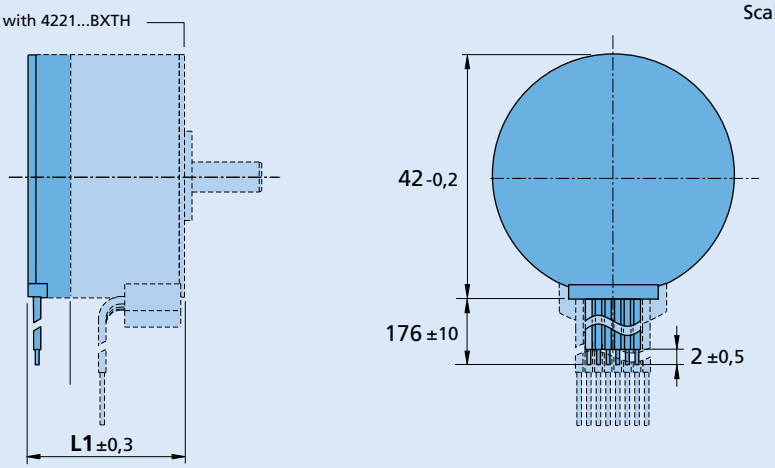


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Dimensional drawing C

Example of combination with 4221...BXTH

Scale reduced



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